

SUPERPOSITION, FOURIER, + TIMBRE

- LINEARITY OF SOUND WAVES + SUPERPOSITIONS
- WAVEFORMS AND TIMBRE
- HARMONICS AND FOURIER DECOMPOSITION
 - OVERTONES

LINEARITY AND SUPERPOSITION

- FOR THIS COURSE, SOUND WAVES ARE IN THE "LINEAR REGIME"

⇒ IF SEVERAL THINGS PRODUCE SOUND, THEN THE TOTAL AIR PRESSURE, VELOCITY, DISPLACEMENT, ETC, IS JUST THE SUM OF THE PRESSURES, VELOCITIES, DISPLACEMENTS, ETC FROM EACH THING, E.G.:

EQUILIBRIUM/AVERAGE/ATMOSPHERIC PRESSURE (CONSTANT)

$$P(x, y, z, t) = \bar{P} + \overset{\downarrow \text{SOURCE 1}}{P_1(x, y, z, t)} + \overset{\downarrow \text{SOURCE 2}}{P_2(x, y, z, t)} + \overset{\downarrow \text{SOURCE 3}}{P_3(x, y, z, t)} + \dots$$

TOTAL PRESSURE AT LOCATION (x, y, z) & TIME t

PRESSURE WAVES FROM ALL SOURCES (RECALL THIS IS JUST " ΔP ")

ENTHUSIASTS: PLUG THIS INTO THE WAVE EQUATION TO SEE THAT IT REMAINS VALID (AS LONG AS EACH P_i IS A SOLUTION ON ITS OWN!)

- SIMILARLY THE VELOCITY

$$\vec{v} = \vec{\bar{v}} + \vec{v}_1 + \vec{v}_2 + \dots$$

↑
AVERAGE WIND

- AND DISPLACEMENT

$$\chi = \bar{\chi} + \chi_1 + \chi_2 + \dots$$

↑
JUST AN ARBITRARY OFFSET (USUALLY SET TO ZERO)

- BASICALLY, EACH WAVE PROPAGATES INDEPENDENTLY OF ALL OTHER WAVES.

EXAMPLE:

AN ORCHESTRAL SOUND WAVE CAN BE DESCRIBED AS A SUM OF SIMPLE SOUND WAVES (FROM EACH INSTRUMENT)

- THIS APPROXIMATION IS VERY ACCURATE AS LONG AS

SPEED OF AIR CHUNKS → $v \ll c \approx 344 \text{ m/s}$

SLOW AIR OSCILLATIONS
(PAIN THRESHOLD $\sim 0.1 \text{ m/s}$)

$$\Delta P \equiv P - \bar{P} \ll \bar{P}$$

SMALL PRESSURE CHANGES
($\sim \text{ppm}$ FOR TRAFFIC)

BASICALLY "SMALL AMPLITUDE"

- VIOLATED BY THINGS LIKE VIOLENT EXPLOSIONS (WE WILL IGNORE THIS)

APPLICATION TO PITCH & TIMBRE

- MOTIVATION: DIFFERENT WAVEFORMS HAVE DIFFERENT TIMBRES.

DEMO

IN SOUND EDITOR, GENERATE

- SINE
- TRIANGLE
- SQUARE
- SAWTOOTH

ANYONE WANT TO TRY TO IMITATE THESE?

PLAY & NOTE THE RELATIONSHIP BETWEEN "BITE" OR "SHARPNESS" OF SOUND AND THE PRESENCE OF SHARP FEATURES IN THE WAVE. SAME PITCH, DIFFERENT TIMBRE!

- SIMILAR WITH SINGING:

RECORD "UUU", "EEEE", "RRR"... & INSPECT

- HOW DO WE RECOGNIZE VOWELS?

- WHAT IS REALLY GOING ON HERE? HOW CAN WE QUANTIFY THE TIMBRE?

PURE TONES

- TO GET A FOOTHOLD, CONSIDER A "PURE" TONE, I.E., A SINGLE SINUSOID:

$$P - \bar{P} \equiv \Delta P = \underbrace{\Delta P_0}_{\text{CONSTANT AMPLITUDE}} \sin\left(2\pi f t + \underbrace{\phi_0}_{\text{CONSTANT PHASE (SHIFTS WHOLE WAVE ALONG TIME AXIS)}}\right) = \Delta P_0 \sin\left(2\pi \frac{t}{T} + \phi_0\right)$$

RELATIVE PRESSURE (points to $P - \bar{P}$)
FREQUENCY (points to f)
SINE FUNCTION REPEATS EVERY 2π (points to 2π)
OUR SINE REPEATS EVERY PERIOD $T = \frac{1}{f}$ (points to $\frac{t}{T}$)

- PURE TONES ARE A "BUILDING BLOCK" OF MORE COMPLEX WAVES, AS WE SHALL NOW SEE!

AFTER
WED 9-11

SUPERPOSITIONS OF PURE TONES

DEMO

(PHET) FOURIER: MAKING WAVES

- MAXIMIZE OVERALL VOLUME ON RIGHT
- SET "FUNCTION OF:" TO "TIME (t)"
- DISCUSS AXES & PERIODICITY (ASSUME IT REPEATS)

10pt

Q

WHAT IS THE FREQUENCY OF THIS FUNCTION?

- PLAY + STOP
- SWITCH "SERIES:" TO "COSINE"

5pt

Q WHAT PITCH IS THIS?

10pt

Q WILL THIS TONE SOUND DIFFERENT?

10pt

Q WHAT IS THE FREQUENCY OF THIS WAVE?

10pt

Q IF WE ADD THIS TO A_1 , HOW OFTEN WILL THE RESULTING WAVE REPEAT?

- INCREASE A_2 + SEE
- PLAY TONE. DOES IT SOUND MORE LIKE A_2 OR A_3 ?

CONVENTIONAL
WISDOM:
(YRMV)

BRAIN SEEMS BIASED TOWARD
"PITCH" = LOWEST PRESENT FREQUENCY.

- ADJUST A_3 UNTIL IT LOOKS MORE TRIANGULAR
- ADJUST A_3 UNTIL IT LOOKS MORE SQUARE

⇒ YOU CAN MAKE DIFFERENT WAVEFORMS
WITH DIFFERENT SUPERPOSITIONS

- SET "EQUATION:" TO "f" TO SHOW MATH.

- IT TURNS OUT THAT ANY WAVEFORM WITH PERIOD T CAN BE WRITTEN AS

$$P(t) = \bar{P} + P_1 \cos(2\pi f_1 t + \phi_1) + P_2 \cos(2\pi f_2 t + \phi_2) + \dots$$

WITH $f_n = n f_1 = \frac{n}{T}$ $\leftarrow 1, 2, 3 \dots$ FREQUENCY-DEPENDENT AMPLITUDE AND PHASE

"FOURIER SERIES"

FREQUENCIES ARE "HARMONICS" OF f_1

- VISUAL:



- THE ACT OF FINDING P_n 's + ϕ_n 's FROM $P(t)$ IS CALLED A "FOURIER TRANSFORM"

ENTHUSIASTS:

- OR, USING "TRIGONOMETRY," ($\cos(A+B) = \cos A \cos B - \sin A \sin B$)

$$A_n \cos(2\pi f_n t + \phi_n) = A_n (\cos(2\pi f_n t) \cos \phi_n - \sin(2\pi f_n t) \sin \phi_n)$$
$$= \underbrace{A_n \cos \phi_n}_{\equiv X_n} \cos(2\pi f_n t) - \underbrace{A_n \sin \phi_n}_{\equiv Y_n} \sin(2\pi f_n t)$$

TRADITIONAL
FORM (EASIER
TO CALCULATE)

$\bar{P} \rightarrow X_0$ (NOT "POSITION," UNITS P_n)

$$\Rightarrow P(t) = X_0 + X_1 \cos(2\pi f_1 t) + X_2 \cos(2\pi f_2 t) + \dots + Y_1 \sin(2\pi f_1 t) + Y_2 \sin(2\pi f_2 t) + \dots$$

NOTICE $X_n^2 + Y_n^2 = A_n^2$

- NOTE IF YOU KNOW X_n + Y_n , YOU KNOW P_n + ϕ_n

10pt

Q IS $P(t)$ WRITTEN THIS WAY GUARANTEED TO REPEAT? (WHEN?)

DEMO

PLOT FFT OF SOME SOUNDS IN PYTHON
OR AUDACITY

(NEEDED LATER)

10pt

Q WHY ARE THERE SO MANY FOURIER COEFFICIENTS?
(WHY ARE THEY SO DENSE?)

ENTHUSIASTS

TO CALCULATE X_n , WE ASK "HOW MUCH DOES $P(t)$ OVERLAP WITH $\cos(2\pi f_n t)$?" FOURIER'S TRICK:

$$\int_0^T P(t) \cos(2\pi f_n t) dt = \int_0^T [\bar{P} + X_1 \cos(2\pi f_1 t) + \dots + \underbrace{X_n \cos(2\pi f_n t)}_{\text{ONLY ONE THAT DOES NOT AVERAGE TO ZERO! } \langle \cos^2(\dots) \rangle = \frac{1}{2}} + \dots] \cos(2\pi f_n t) dt$$

COMPUTER DOES THESE NUMERICALLY ("FFT") $= \frac{T}{2} X_n$ ← SOLVE FOR X_n ☺

BACK TO SHOW!

10pt
Q

CAN YOU TELL THE DIFFERENCE BETWEEN POSITIVE A_3 AND NEGATIVE A_3 ?

- BUT SWITCHING THE SIGN OF THE HARMONIC CAN CHANGE THE WAVEFORM FROM SQUARE-LIKE TO TRIANGLE-LIKE. SHOULDN'T THEY SOUND DIFFERENT?

10pt
Q

WHY DO SQUARE WAVES SOUND DIFFERENT FROM TRIANGLE WAVES?

TAKE-HOME MESSAGE:

A WAVEFORM'S TIMBRE COMES FROM THE RELATIVE AMPLITUDES OF ITS CONSTITUENT PURE TONES

DEMO

- FFT 192 Hz CLICKS, THEN SWITCH TO LOG X
- FFT A CHROMATIC SCALE THEN SWITCH BACK TO LINEAR.
- REVISIT

WHAT IS THE PEAK SPACING?

- USUALLY BEST TO USE LINEAR f-SCALE FOR HARMONICS (e.g. DECOMPOSING A TONE) AND LOG FOR MUSICAL STRUCTURE

WE WILL COME BACK
TO THIS!



OVERTONES CAN BE ANY TONES WITH f ABOVE THE FUNDAMENTAL, NOT JUST HARMONICS
- THEY MAY HAVE FREQUENCY THAT PREVENTS THE WAVEFORM FROM EVER
REPEATING. (IRRATIONAL RATIO)
- HUMANS SEEM TO LIKE WHEN IT CAUSES A REPEAT EVERY FEW
FUNDAMENTAL PERIODS. (THESE ARE THE "PLEASANT CHORDS")

SEE ASSIGNMENT 2!

SUMMARY

- SOUND WAVES (AS WE CARE TO KNOW THEM) ARE "LINEAR"

EQUILIBRIUM/AVERAGE/ATMOSPHERIC PRESSURE (CONSTANT)

$$\Rightarrow \underbrace{P(x, y, z, t)}_{\text{TOTAL PRESURE AT LOCATION (x, y, z) \& TIME t}} = \underbrace{\bar{P} + P_1(x, y, z, t) + P_2(x, y, z, t) + P_3(x, y, z, t) + \dots}_{\text{PRESSURE WAVES FROM ALL SOURCES (RECALL THIS IS JUST "DP")}}$$

TOTAL PRESURE
AT LOCATION
(x, y, z) & TIME t

PRESSURE WAVES FROM ALL SOURCES
(RECALL THIS IS JUST "DP")

- ALL PERIODIC WAVES (PERIOD T) CAN BE WRITTEN AS A "FOURIER SERIES"

$$P(t) = \bar{P} + A_1 \cos(2\pi f_1 t + \phi_1) + A_2 \cos(2\pi f_2 t + \phi_2) + \dots$$

PHASES NOT "HEARD" ABOVE $f_n \sim 20 \text{ Hz}$.

THESE A_n 'S ARE THE AMPLITUDES OF "HARMONIC"
FREQUENCY $f_n = \frac{n}{T}$.

- PLOTTING A_n VS f_n CREATES A "SPECTRUM" FOR THE TONE,
WHICH DETERMINES ITS TIMBRE.